Project Semester August-January 2025

DATA SCIENCE MINOR PROJECT REPORT

ON

Real-Time Weather Forecast and Air Quality Monitoring Dashboard using Power Bi

DATA ANALYTICS WITH POWER BI

COURSE CODE: INT374

B. TECH COMPUTER SCIENCE AND ENGINEERING

LOVELY PROFESSIONAL UNIVERSITY

PHAGWARA, PUNJAB

PROJECT SUBMITTED BY:

Tanvi (12315499)

Section: K23SB

Roll No.: 50

PROJECT SUBMITTED TO:

Ms. Sadaf Qureshi

DECLARATION

I, **Tanvi**, student of B.Tech – Computer Science and Engineering (Section K23SB) at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report titled:

"Real-Time Weather Forecast and Air Quality Monitoring Dashboard using Power Bi''

is based on my own intensive work and is genuine. The content of this report has not been submitted to any other university or institution for the award of any degree or diploma.

Date: 15-12-2025
Registration No.: 12315499
Name: Tanvi

CERTIFICATE

This is to certify that **Ms. Tanvi,** bearing Registration No. **12315499**, has successfully completed the **INT374** – Introduction to Data Management project titled:

Real-Time Weather Forecast and Air Quality Monitoring Dashboard using Power Bi

under my guidance and supervision. To the best of my knowledge, the present work is the result of her original development, effort, and study. This project has been carried out as a part of the curriculum prescribed by Lovely Professional University, Phagwara for the Project Semester **August–January 2025.**

SignatureandNameoftheSupervisor
Ms.Sadaf Qureshi

ACKNOWLEDGEMENT

I sincerely thank Ms. Sadaf Qureshi , for her guidance and support throughout this project. I also thank the faculty of the CSE Department at Lovely Professional University for providing the necessary resources and assistance.

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1. INTRODUCTION

Environmental conditions such as weather and air quality have a direct impact on public health, transportation, agriculture, and daily human activities. With increasing urbanization and pollution levels, access to real-time and accurate weather and air quality information has become essential.

This project focuses on the development of an interactive Weather and Air Quality Analytics Dashboard using Power BI, powered by live data from the OpenWeather API. The dashboard provides real-time insights into weather conditions and air pollution levels across multiple cities, enabling users to understand environmental trends and make informed decisions.

2.SOURCE OF DATASET

OpenWeatherMap Public APIs (Weather & Air Pollution) – Data fetched in real time using latitude and longitude for Indian capital cities via REST API integration.

3.DATASET PREPROCESSING

Data preprocessing is a crucial step to ensure the accuracy, consistency, and usability of weather and air quality data before visualization and analysis. The raw data obtained from the OpenWeather API is provided in nested JSON format and requires multiple transformation steps to convert it into a structured, analysis-ready dataset.

1. Data Extraction

Weather and air pollution data were fetched using the OpenWeather API through Power BI's Web connector. The API returns real-time atmospheric parameters, pollutant concentrations, and Air Quality Index (AQI) values in JSON format for multiple cities.

2. Data Transformation

The raw JSON response contained deeply nested records and lists. Power Query was used to:

- Expand nested JSON fields such as current, main, and components
- Extract pollutant concentrations including PM2.5, PM10, CO, NO_2, SO_2, and O_3
- Remove unnecessary attributes like latitude, longitude, and redundant weather objects

This step converted the semi-structured JSON data into a clean tabular format.

3. Date and Time Standardization

The API provides timestamps in UNIX epoch format. These values were converted into human-readable date and time using Power Query, and adjusted to the local time zone for accurate temporal analysis.

4. Data Cleaning

Data cleaning was performed to ensure reliability:

- Removed irrelevant columns to reduce dataset size
- Ensured numeric data types for all pollutant concentration fields
- Standardized column names for consistency and ease of use
- Handled null and missing values to prevent calculation errors

5. AQI Calculation

Although OpenWeather provides a basic AQI scale (1–5), a more accurate AQI value (0–500 scale) was computed using U.S. EPA standard formulas. Individual AQI values were calculated for each pollutant using defined breakpoint ranges. The final AQI for each city was derived as the maximum AQI value among all pollutants.

6. AQI Categorization

The calculated AQI values were categorized into standard air quality levels:

- Good
- Moderate
- Unhealthy for Sensitive Groups
- Unhealthy
- Very Unhealthy
- Hazardous

This classification enables easy interpretation of air quality conditions for end users.

4.ANALYSIS ON DATASETS:

Objective-wise Analysis and Outcomes

Objective 1: To collect real-time weather and air quality data using OpenWeather API

Outcome / Analysis:
Real-time weather and air pollution data were successfully retrieved for multiple cities using OpenWeather API. The API provided essential atmospheric parameters such as temperature, humidity, wind speed, and pollutant concentrations including PM2.5, PM10, CO, NO_2, SO_2, and O_3. This ensured the dataset remained current and relevant for environmental monitoring.

Objective 2: To preprocess and transform raw JSON data into a structured format

Outcome / Analysis:
The raw API data, received in nested JSON format, was transformed using Power Query into a clean and structured tabular dataset. Irrelevant fields were removed, nested records were expanded, and data types were standardized. This preprocessing improved data quality and prepared the dataset for accurate analysis and visualization.

Objective 3: To calculate accurate AQI values using pollutant concentration data

Outcome / Analysis:
Using U.S. EPA AQI standards, individual AQI values were calculated for each pollutant. The final AQI for each city was derived as the maximum of all pollutant AQI values. This approach provided more precise air quality measurement compared to the default AQI scale provided by the API.

Objective 4: To categorize air quality levels based on AQI standards

Outcome / Analysis:
Calculated AQI values were classified into standard categories such as Good, Moderate, Unhealthy for Sensitive Groups, Unhealthy, Very Unhealthy, and Hazardous. This categorization enabled easy interpretation of air quality conditions and supported health-based insights for different user groups.

Objective 5: To analyze pollutant dominance and its impact on air quality

Outcome / Analysis:
Analysis revealed that particulate matter (PM2.5 and PM10) had the most significant impact on overall AQI across most cities. Gaseous pollutants such as NO_2 and CO showed moderate influence, especially in urban and high-traffic regions. This helped identify key contributors to air pollution.

Objective 6: To study the relationship between weather conditions and AQI

Outcome / Analysis:
Weather parameters such as wind speed, humidity, and rainfall were analyzed against AQI values. Higher wind speeds were generally associated with lower AQI due to pollutant dispersion, while stagnant weather conditions resulted in higher pollution levels. Rainfall events showed a temporary reduction in particulate matter.

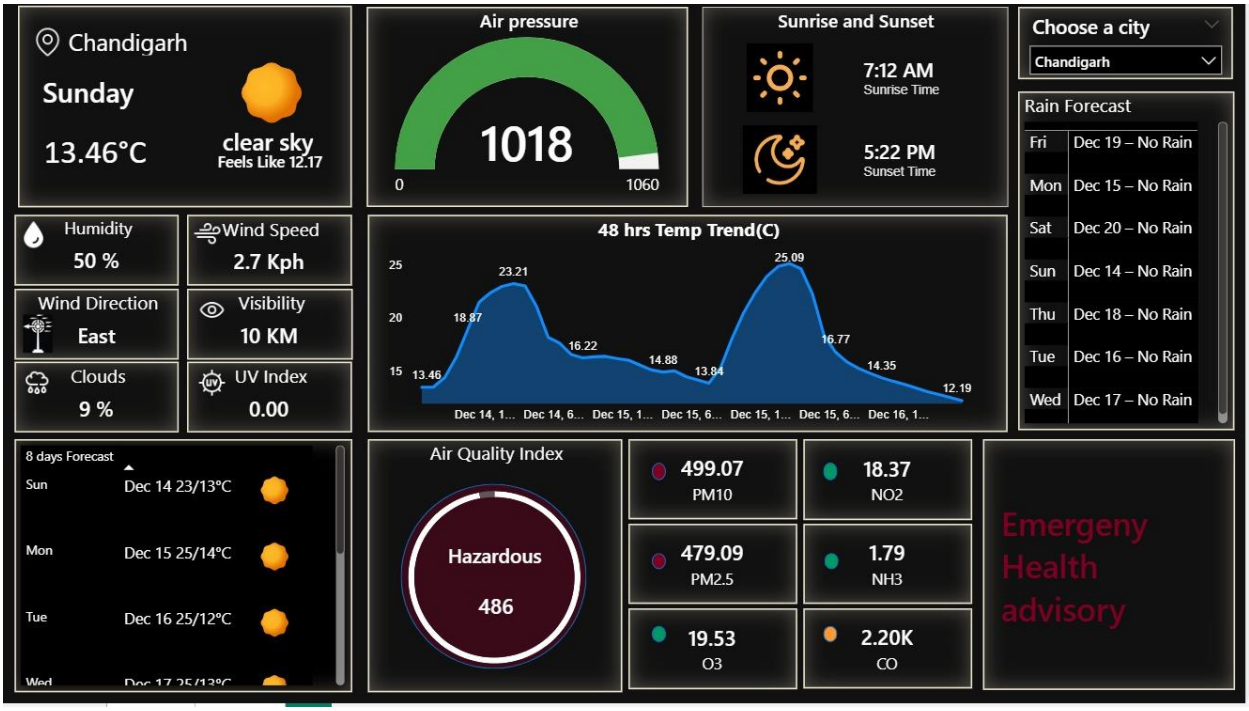
Objective 7: To compare air quality across different cities

Outcome / Analysis:
City-wise comparisons highlighted significant spatial variation in air quality. Metropolitan and industrial cities recorded consistently higher AQI values, whereas coastal and hilly regions exhibited comparatively better air quality. This geographic analysis provided location-specific environmental insights.

Objective 8: To visualize weather and AQI insights using Power BI dashboards

Outcome / Analysis:
Interactive Power BI dashboards were developed using gauges, cards, charts, and slicers. The AQI gauge provided instant air quality assessment, while trend charts and pollutant visuals enabled deeper analysis. These visualizations transformed complex data into easily understandable insights for users.

DASHBOARD



6. Conclusion

The Weather Forecast and Air Quality Monitoring Dashboard developed using Power BI and OpenWeather API successfully transforms real-time environmental data into meaningful and interactive insights. Through effective data preprocessing, accurate AQI computation using EPA standards, and dynamic visualization techniques, the dashboard provides a comprehensive view of weather conditions and air pollution levels across multiple cities.

The project demonstrates the practical application of APIs, data transformation using Power Query, and data modeling in Power BI. The AQI gauge and pollutant analysis enable users to quickly assess air quality and potential health risks. Overall, the dashboard serves as an efficient decision-support tool for environmental awareness, public health interpretation, and data-driven analysis.

7. Future Scope

The project can be further enhanced in the following ways:

- **Predictive AQI Forecasting**
  Machine learning models can be integrated to predict future AQI levels based on historical pollution and weather data.

- **Real-Time Alerts and Notifications**
  Automated alerts can be generated for hazardous AQI levels to warn users and sensitive groups.

- **Expanded Geographic Coverage**
  The dashboard can be scaled to include more cities, states, or global regions.

- **Health Recommendation Integration**
  Personalized health advisories can be displayed based on AQI categories and user profiles.

- **Mobile and Web Deployment**
  The dashboard can be published to Power BI Service and embedded into web or mobile applications for wider accessibility.

- **IoT Sensor Integration**
  Integration with real-time IoT air quality sensors can improve accuracy and reliability.

- **Advanced Visual Analytics**
  Heat maps, animated time-series visuals, and AI-powered insights can enhance data interpretation.

8. References

- OpenWeather API Documentation
  https://openweathermap.org/api
- OpenWeather Air Pollution API
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  https://www.who.int/health-topics/air-pollution
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